The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 2 MODULE, AUTUMN SEMESTER 2017-2018

OPERATING SYSTEMS AND CONCURRENCY
COMP 2035 (G52OSC)

Time allowed TWO Hours

---

*Candidates must **NOT** start writing their answers until told to do so*

Answer THREE questions out of FOUR

Only silent, self-contained calculators with a single line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Operating Systems Structure, Memory Management and File Management (25 Marks)

(a) Explain the structure and the main advantages of using a microkernel approach for designing an Operating System.

[6 Marks]

(b) Describe any one file system implementation that uses linked lists. List any two (2) advantages and one (1) disadvantage of this method.

[4 Marks]

(c) Briefly explain the purpose of *symbolic linking* of files? [2 Marks]

(d) A memory manager allocates memory in fixed size chunks of 2000 bytes to each process. The current allocation is as shown in the Table 1.

Process	Size
P1	1000 bytes
P2	2000 bytes
P3	300 bytes
P4	10 bytes

Table 1.

What is the total amount memory wasted with the above allocation due to internal fragmentation?

[3 Marks]

(e) Briefly describe how you could implement the *LRU* page replacement algorithm in hardware. Is this algorithm the best in all cases? Why or why not?

[6 Marks]

(f) On a system with paging, a process cannot access memory that it does not own; why? How could the operating system allow access to other memory?

[4 Marks]

2. Process Scheduling (25 Marks)

(a) With the help of examples, describe the following scheduling algorithms.

(i) Highest Response Ratio Next (HRRN), [2 Marks]

(ii) Shortest Job First. [4 Marks]

(b) Round Robin process scheduling policy is said to favour CPU bound processes over I/O bound processes. Explain why this may be the case?

[2 Marks]

(c) Explain under what circumstances would a round robin algorithm degrade to a FCFS algorithm.

[2 Marks]

(d) With help of example, briefly describe the Multilevel Feedback Queue Scheduling algorithm.

[4 Marks]

(e) What advantage is there in having different time-quantum sizes on different levels of a multilevel queuing system?

[3 Marks]

(f) How would you prevent starvation in a multi-level queue scheduling algorithm?

[2 Marks]

(g) What scheduling policy would you normally choose for:

(i) A time sharing system

(ii) A batch system [2 Marks]

(h) What is a real-time system? Discuss the impact of a real-time system on the design of operating systems.

[4 Marks]

3. Deadlock, Device Management and Process synchronisation (25 Marks)

(a) Dijkstra's Banker's algorithm has a number of weaknesses that preclude its effective use in real systems. List any four (4) of their weaknesses.
[4 Marks]

(b) Explain what is meant by the term *deadlock*. [2 Marks]

(c) The fact that a state is *unsafe* does not necessarily imply that the system will be in *deadlock*. Explain why this is true.
[2 Marks]

(d) How does Direct Memory Access (DMA) work and how is it useful? [6 Marks]

(e) Consider the following snapshot of a system in the Table 2. There are no current outstanding queued unsatisfied requests.

Resource Usage

Processes	Current Allocation				Maximum Demand				Resources Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P1	0	0	1	2	0	0	1	2	2	1	0	0
P2	2	0	0	0	2	7	5	0				
P3	0	0	3	4	6	6	5	6				
P4	2	3	5	4	4	3	5	6				
P5	0	3	3	2	0	6	5	2				

Table 2.

Answer the following questions using banker's algorithm.

(i) Calculate the number of resources needed to satisfy the maximum demand of the processes p1, p2, p3, p4, p5 and p6.
[2 Marks]

(ii) Is this system in a safe state? [2 Marks]

(iii) If a request from process P3 arrives for (0, 1, 0, 0), can that request be granted immediately?
[3 Marks]

(f) What reasonable safeguards might be built into an operating system to prevent processes from waiting indefinitely for an event?
[4 Marks]

4. Process synchronisation, Mutual Exclusion and semaphore (25 Marks)

(a) Write any three problems that may arise when disabling the interrupts during the execution of a long critical section.

[3 Marks]

(b) Why is it considerably more difficult to test, debug, and prove program correctness for concurrent programs than for sequential programs?

[5 Marks]

(c) Briefly describe the term *busy waiting*? Is it always wasteful? [4 Marks]

(d) Propose and describe a synchronisation tool that does not require busy waiting.

[5 Marks]

(e) Two phased locking by a database system is similar to requesting all resources needed in advance. Why is this strategy not applicable in general?

[3 Marks]

(f) Consider the following pseudo-code about the critical-section problem of two-threads. Note that the `threadNumber` is global variable, to announce which thread may enter the critical region.

```
int threadNumber = 0;

void ThreadZero()
{
  while (TRUE) do
  {
      while (threadNumber == 1) do {} //spin-wait
      CriticalRegionZero;
      threadNumber = 1;
      OtherStuffZero;
  }
}

void ThreadOne()
{
  while (TRUE) do
  {
      while (threadNumber == 0) do {} //spin-wait
      CriticalRegionOne;
      threadNumber = 0;
      OtherStuffOne;
  }
}
```

Does this version of code enforce the conditions for mutual exclusion? Explain why or why not.

[5 Marks]